IoT Use Case : Asset Tracking & Management

Modern Asset-tracking System

Tracking automation
Track real-time data on a dashboard (location, condition, and performance)

Inventory management
Feed item details into supply chain algorithms

Alerts and notifications
Alerts for delayed or deviated movement during transfer

Chain of custody
Various use cases require a complete trace of the asset location.

Vast application
from cheap SKU in shop / warehouse to expensive items

IoT Use Case : Asset Tracking & Management

The Manual Era: When Everything Was Slow and Human-Driven

Manual tracking created blind spots across the supply chain — slow, error-prone, and reactive.

Manual Asset Tracking

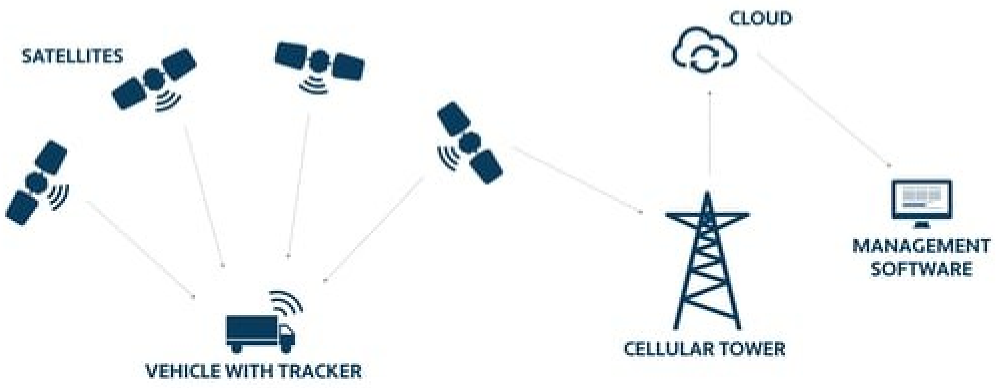
- Spreadsheets, barcode scans, paper logs, manual updates
- Human-driven processes with limited automation

Key Challenges

- Outdated data, Human errors
- Frequent misplace or delayed,
- No real-time visibility, no automated alerts,
- no reliable chain of custody.

IoT Use Case : Asset Tracking & Management

The IoT Era: When Devices Learned to Talk



IoT improved visibility but lacked real-time decision-making and learning.

IOT Enabled Asset Tracking

- Sensors now track assets automatically.
- Real time dashboards , real time data
- Accurate inventory
- Automated inventory with Indoor & outdoor tracking
- Better visibility across warehouses and logistics

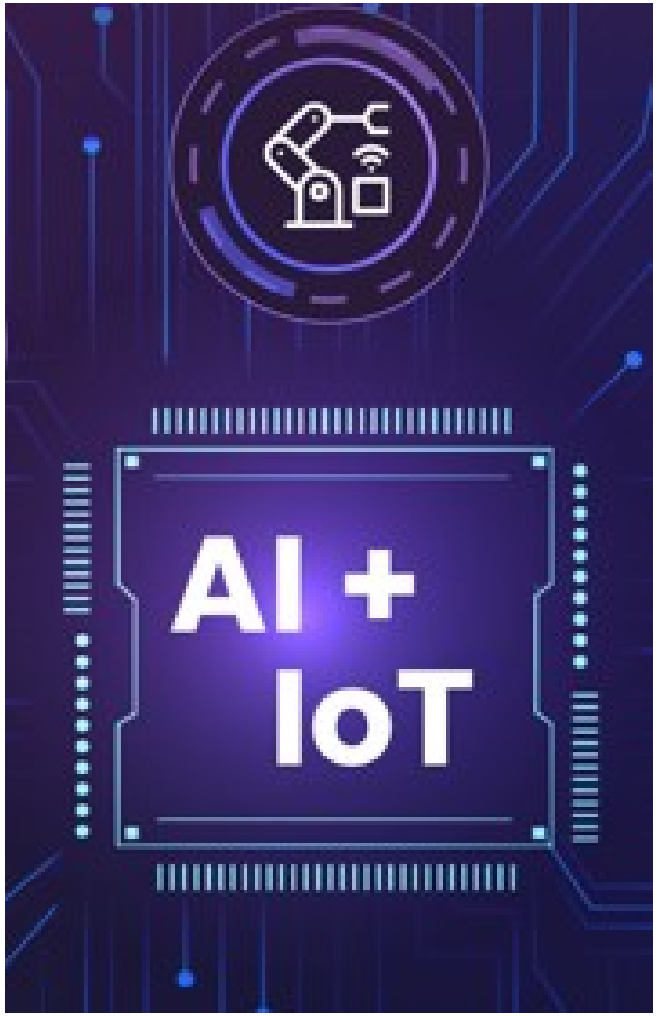
Limitations

- IoT gave visibility, but not intelligence.
- IoT devices collected data and pushed it to the cloud
- Slow reactions as everything depended on cloud processing

IoT Use Case : Asset Tracking & Management

AIOT Era : Intelligent Devices

AI models became lightweight, edge became powerful, cloud AI became scalable and affordable



AIOT - intelligent asset-tracking

- AIOT combines IoT's sensing power with AI's decision-making ability.
- Edge runs lightweight AI models that can detect anomalies instantly.
- Cloud analyzes long-term patterns, retrains models, and sends smarter versions back to the edge
- Continuously learning system that becomes more accurate over time

Advantages

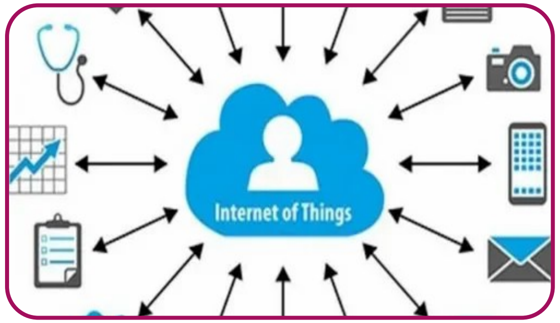
- reduces losses, improves forecasting, strengthens chain-of-custody tracking, and works seamlessly indoors, outdoors, and across borders

AIOT is still evolving

- **More autonomous future -** systems that self-optimize, self-heal, and collaborate across entire supply chains

IoT Use Case : Asset Tracking & Management

Technology evolution : from manual to IoT to AIOT



Manual Era

- Limited technology - Basic computing. Spreadsheets, barcode scanners, and human-driven processes
- **Key Constraints** : Affordable sensors & communication

The rise of IoT

- sensors became cheap,
- wireless became reliable,
- Rise of cloud computing
- **Key Constraint** : devices could talk, but couldn't think.

Birth to AIOT

- Lightweight AI for small processors.
- Powerful Edge devices to analyze data locally.
- Cloud AI enabling cloud-driven learning

Future

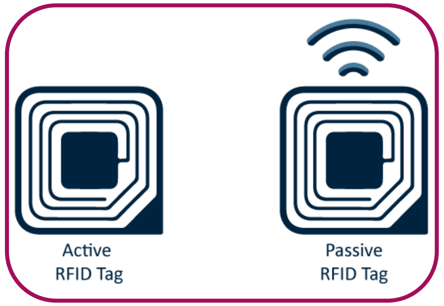
- ultra-low-power AI chips.
- 5G / 6G networks.
- Digital twins
- Autonomous logistics
- generative AI
- self-managing, self-optimizing, and deeply predictive

IoT Use Case : Asset Tracking & Management

Som popular tracking Devices

barcodes and
QR codes

BLE beacons



Wi-Fi-based
tags

GPRS Cell ID

GPS tags

Ultra-Wideband
tags

AirFinder

AirFinder